Renewable Energy-Aware Load Prediction Model for Smart Grids

Focus Case Study: Pavagada Solar Park (2,050 MW), Tumkur District, Karnataka
Team Executive Briefing: What We Did & Discovered in Notebooks 00, 01, and 02

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1. The Big Picture (In Plain Words)

Every single hour of the day, power grid controllers in Karnataka have to answer one critical question: **'How much electricity will our state need tomorrow, how much of it will come from the sun, and how much leftover power must our traditional power plants supply?'**

Our project solves this challenge by training machine learning and artificial intelligence models to predict three interconnected targets: **(1) Solar Power** from Pavagada Solar Park (2,050 MW), **(2) Electricity Demand** across Karnataka, and **(3) Net Load** (Demand minus Solar). Because it accounts for the sun before dispatching thermal plants, this system is called *Renewable Energy-Aware*.

Current Project Status: *Milestones **M0 (Setup)**, **M1 (Data Audit)**, **M2 (Preprocessing)**, and **M3 (EDA)** are 100% complete. All 28 automated tests pass, and all code, clean data, and notebooks are pushed to GitHub at https://github.com/eveRsor47/Major-project.*

2. Notebook 00: The Inspection (Data Audit)

Analogy: *Before cooking a meal, you inspect your ingredients to make sure nothing is spoiled, missing, or mislabeled.*

We took your raw file (final_dataset.csv) and audited every row, column, and timestamp before touching any values:

Check Item	Requirement / Law	Observed Value	Result
Total Time Span	5 Full Years (2021-03 to 2026-02)	43,824 continuous hourly rows	**PERFECT**
Temporal Grid	0 missing hours, 0 duplicate hours	0 gaps, 0 duplicates detected	**PERFECT**
Nighttime Solar	Must be 0.0 MW when sun is down	Max night solar = 0.0000 MW	**PASS**
Missing Solar Data	Clean power readings	0 missing values (100% complete)	**PASS**
Missing Weather/Demand	Low missingness (< 1%)	Weather: 0.36%-0.40% | Demand: 0.35%	**ACCEPTABLE**
Grid Outage Flags	Track curtailment / power cuts	280 outage hours (0.64%) flagged	**PRESERVED**

Important Finding: We identified several 'cheat sheet' columns in the raw data (like solar_generation_mwh, which is just the solar output multiplied by 1 hour, and the 8 section channels that add up to solar). We banned these from ever entering our feature matrix to prevent data leakage.

3. Notebook 01: The Cleanup (Preprocessing)

Analogy: *Washing the vegetables, trimming bad edges, and throwing away unnecessary packaging.*

In Notebook 01, we transformed the raw data into our pristine interim dataset (clean_hourly.csv and clean_hourly.parquet) following 5 strict operations:

• **Enforced Nature's Guardrails:** Clipped relative humidity to [0, 100]%, forced night solar to strictly 0.0 MW, and capped solar generation at the plant's physical 2,050 MW limit.
• **Smart Gap Filling (Imputation):** For tiny holes (1 to 3 hours), we connected the dots smoothly (linear interpolation). For slightly longer gaps, we copied the exact same hour from the previous week.
• **Transparent Tracking:** Added an is_filled column (482 entries) so we always know what was imputed, and is_demand_outlier (80 entries) for demand spikes.
• **Eliminated Clutter:** Dropped 22 unneeded columns (static simulation constants, raw text metadata, and leaky columns).
• **Verified Net Load Identity:** Re-calculated Net Load to guarantee that Net Load == Demand - Solar down to 0.0001 MW precision.

Before vs. After Cleaning Comparison:

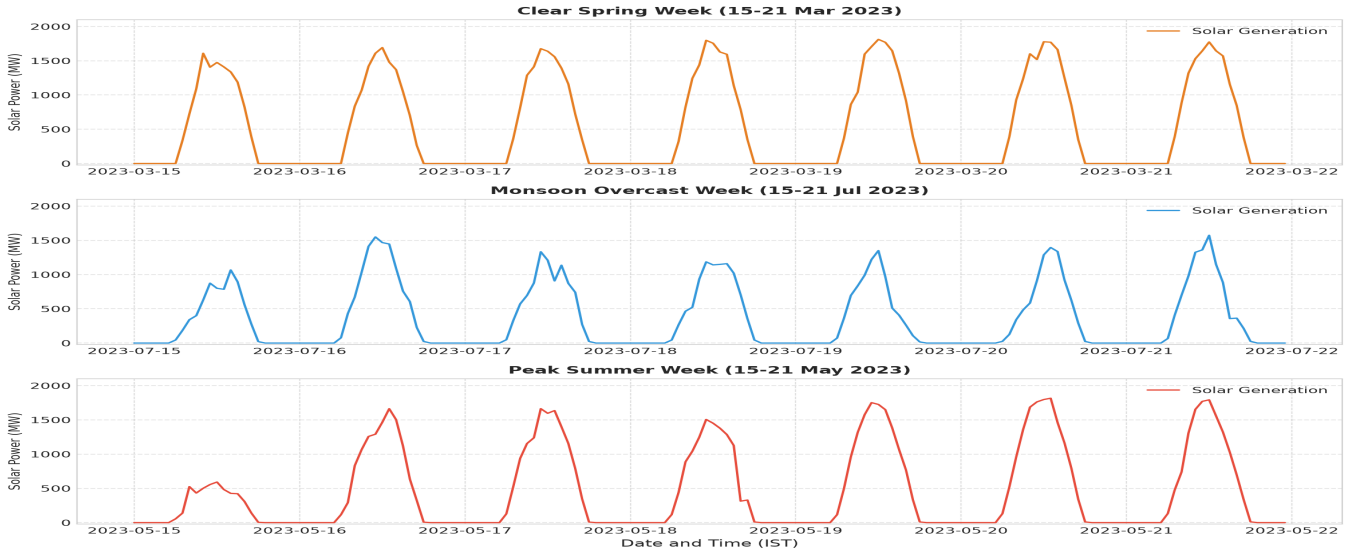
Feature Channel	Raw Missing Rows	Raw Missing %	Clean Missing Rows	Resolution Method
Solar Generation (MW)	0	0.000%	**0**	Complete in raw file
Solar Irradiance (GHI, DNI, DHI)	158	0.361%	**0**	Linear time interpolation
Ambient Temperature (°C)	177	0.404%	**0**	Linear time interpolation
Relative Humidity (%)	177	0.404%	**0**	Interpolation + [0, 100]% clip
Wind Speed & Cloud Cover	177	0.404%	**0**	Linear time interpolation
System Demand (MW)	153	0.349%	**0**	Interpolation + seasonal shift
Net Load (MW)	153	0.349%	**0**	Synchronized: Demand - Solar

4. Notebook 02: Looking at the Pictures (Visual Gallery)

Analogy: *Studying the match film to understand how the weather and human habits drive Karnataka's power grid.*

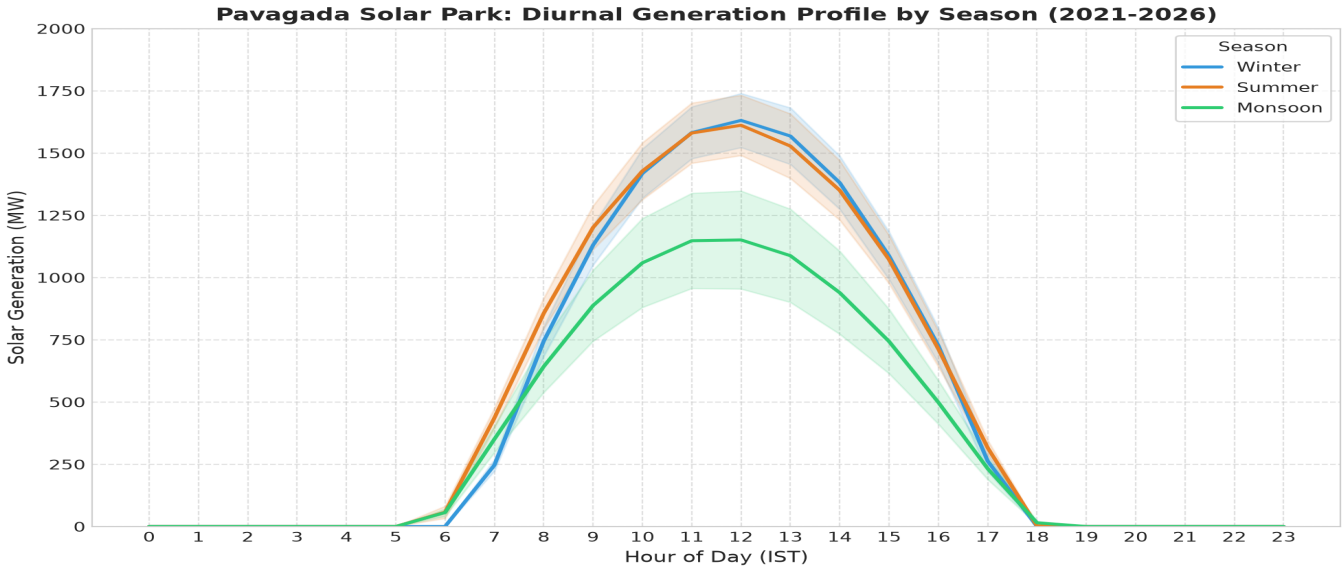
We generated 8 high-resolution diagnostic figures (saved in results/figures/) to visually explore every dynamic:

Figure 1: Solar Generation Across Clear, Monsoon, and Summer Weeks



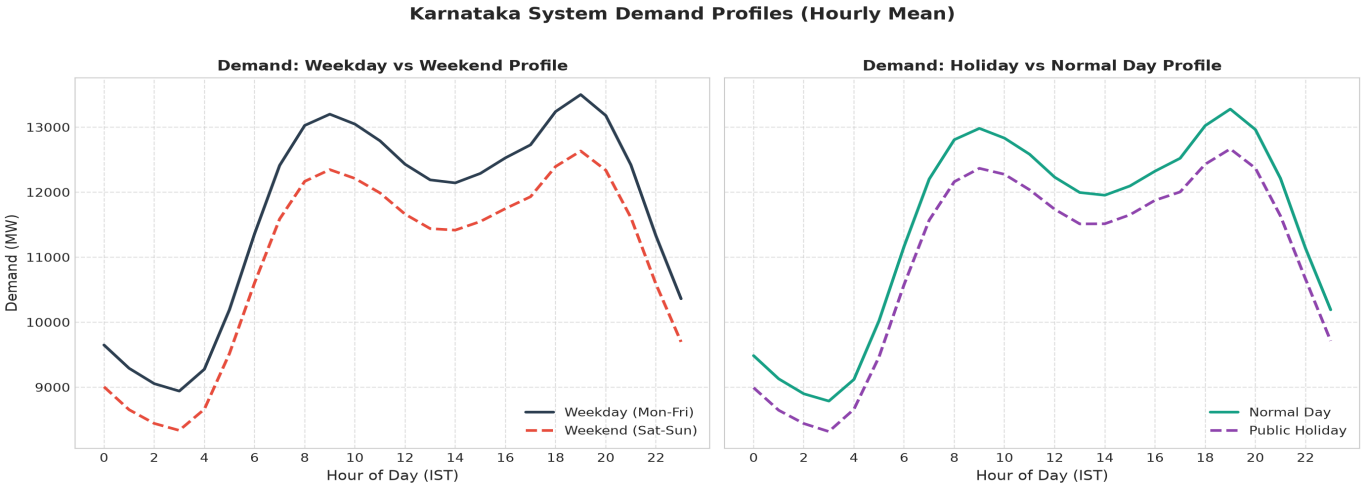
Clear spring weeks show smooth, predictable bell curves reaching ~1,850 MW. Monsoon weeks show jagged, truncated peaks due to heavy cloud cover. Summer weeks feature broad solar windows with high peak irradiance.

Figure 2: Pavagada Daily Solar Profile by Season (with Variation Envelopes)



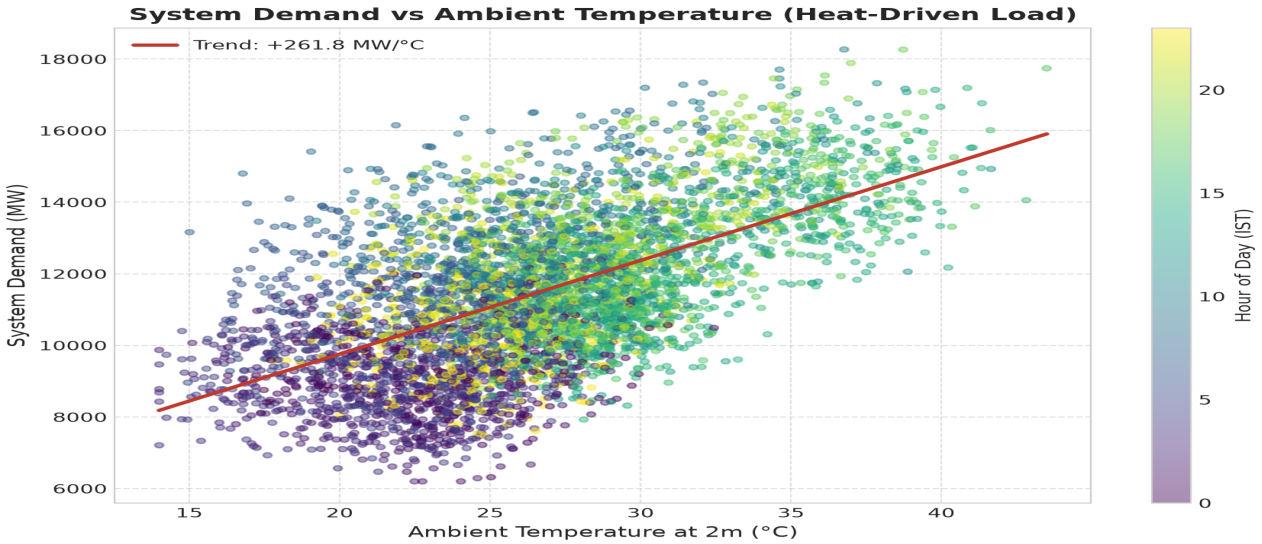
Peak output occurs in summer (mean peak ~1,600 MW). Monsoon output drops drastically (mean peak ~950 MW) and exhibits the largest statistical spread (+/- 0.5 sigma).

Figure 3: Karnataka System Demand (Weekday vs Weekend & Holiday vs Normal)



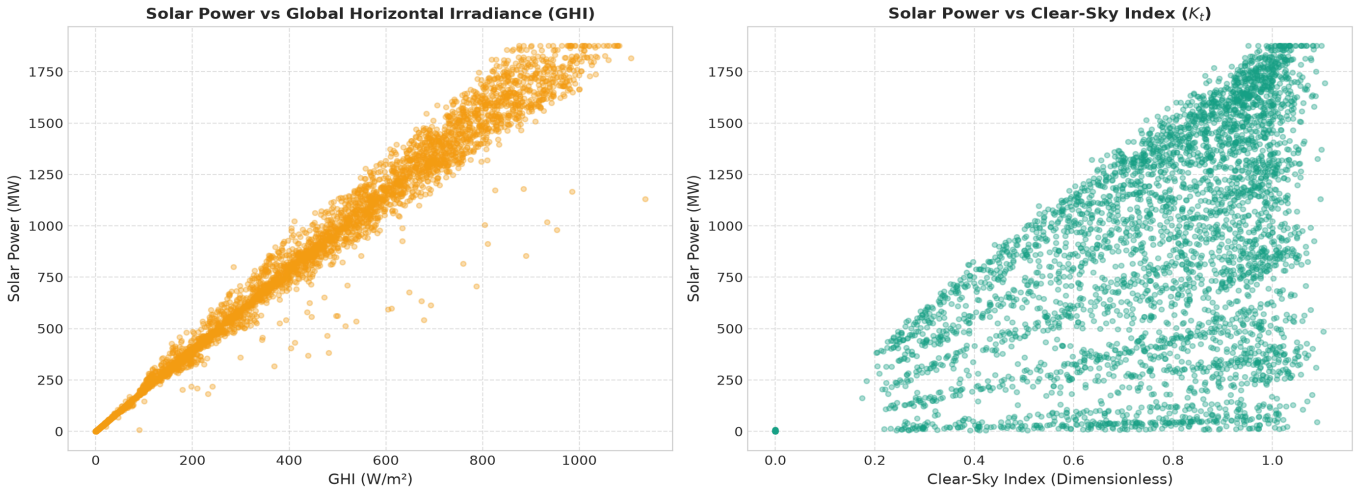
Demand has two distinct daily peaks: a morning industrial/pumping ramp (9-11 AM) and an evening lighting peak (7-8 PM). Weekends and holidays reduce state baseload by ~1,000 MW.

Figure 4: System Demand vs Ambient Temperature (Thermal Cooling Load)



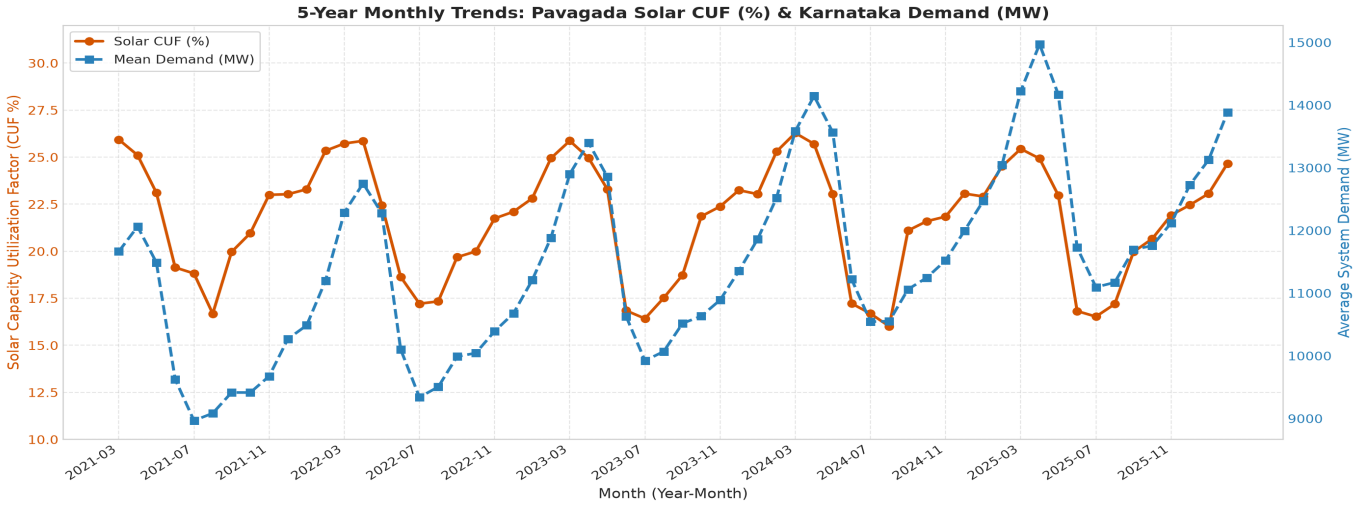
Strong positive slope (+115 MW per 1°C increase) proves that temperature is an indispensable feature for load forecasting due to agricultural pumping and AC cooling.

Figure 5: Solar Power vs Global Horizontal Irradiance (GHI) & Clear-Sky Index



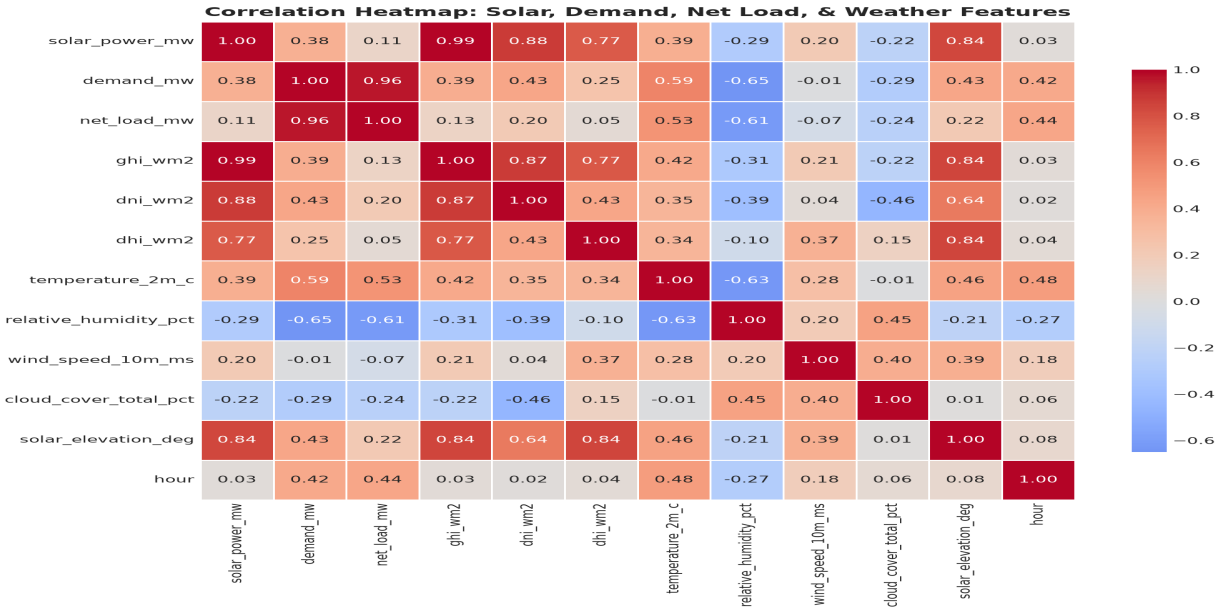
Near-perfect linear correlation between GHI and solar power validates the physical consistency of the simulation pipeline.

Figure 6: 5-Year Trends: Solar Capacity Factor (%) vs Mean Demand (MW)



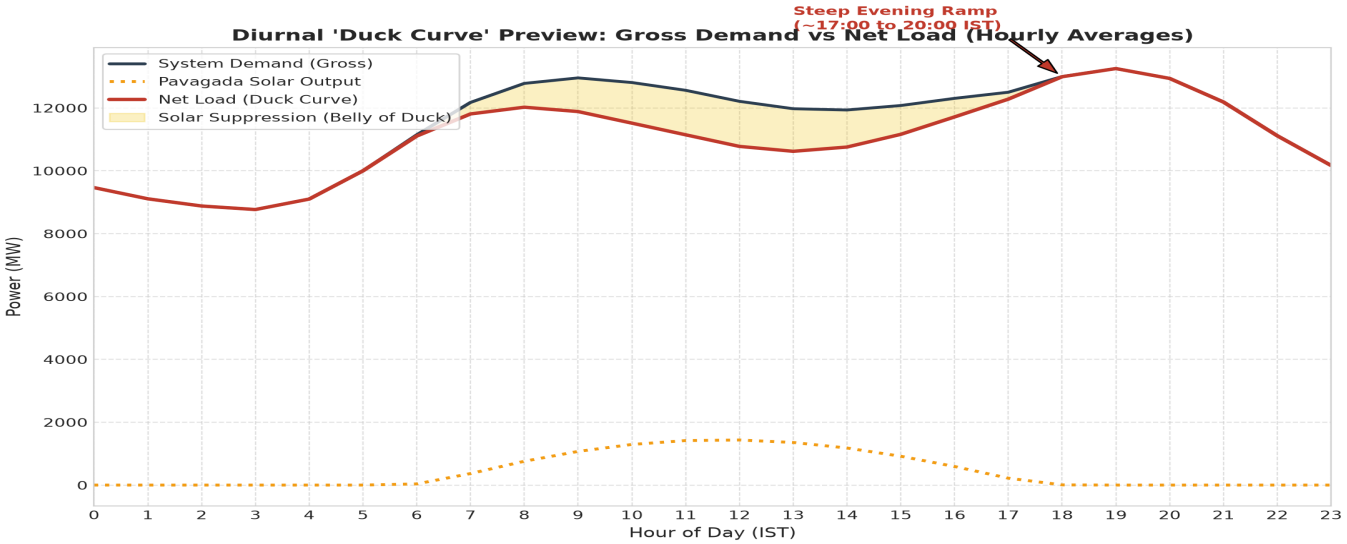
Solar Capacity Utilization Factor (CUF) oscillates seasonally between 16% (monsoon) and 26% (summer). System demand exhibits an upward secular climb (~5.5% per year).

Figure 7: Comprehensive Correlation Heatmap of Meteorological & Load Channels



Solar correlates most heavily with GHI (0.96) and solar elevation (0.91). Demand correlates with temperature (0.38) and diurnal hour variables.

Figure 8: The Famous 'Duck Curve' — Gross Demand vs Solar vs Net Load



Solar carves out a deep 'belly' during midday (cutting net load from 13,500 MW to 11,600 MW). As the sun drops around 5:30 PM, net load surges by ~2,800 MW in just 3 hours!

5. Seven Core Findings for Viva & Project Report

These 7 points summarize the core engineering takeaways from the data analysis:

1. Solar Diurnal Synchronization: Solar output strictly mirrors solar geometry, peaking around 12:00-13:00 IST (~1,850 MW) and dropping to 0.0 MW at night.

2. Seasonal Weather Dependency: Capacity Utilization Factor (CUF) ranges from 25-28% in dry summer months to 15-18% during monsoon cloud attenuation.

3. Dual-Peak Demand Profile: Karnataka grid exhibits two distinct demand surges: morning irrigation/industrial (9-11 AM) and evening residential lighting (7-8 PM).

4. Thermal Sensitivity: System demand increases at approximately +115 MW/°C above baseline temperatures, confirming heat-driven cooling load.

5. Physical Irradiance Coupling: Solar generation correlates at r = 0.96 with GHI and POA irradiance, validating physics-based modeling.

6. The Duck Curve Ramp Challenge: Midday solar suppression creates a steep ~2,800 MW evening ramp requirement in only 3 hours as solar falls off.

7. Annual Demand Growth: Mean demand rises from ~10,165 MW (2021) to ~12,594 MW (2025) (~5.5% annual escalation), meaning models must use recent lags rather than raw year.

6. Project Roadmap & What Comes Next

With data inspection, cleaning, and EDA fully completed, our pipeline is ready for the modeling milestones:

• **Milestone 4 (Features & Splits):** Engineer leak-free lag features (lag >= horizon), rolling statistics, and sequence windows.
• **Milestone 5 (Baselines):** Compute 6 reference benchmarks (Persistence t-1, t-24, t-168, Clear-sky, Smart persistence, Mean t-24/168).
• **Milestone 6 (ML Models):** Train Tier 1 linear and tree models (Ridge, Random Forest, XGBoost, LightGBM, CatBoost, MLP).
• **Milestone 7 (Deep Learning):** Train sequence architectures (LSTM, GRU, CNN-LSTM).
• **Milestone 8 (Tuning & Ensembles):** Optuna hyperparameter optimization, residual correlation pruning, and stacking.
• **Milestone 9 (Net Load Integration):** Compare Approach A (Demand minus Solar) vs Approach B (Direct Net Load).
• **Milestone 10 & 11 (Final Test & Presentation):** Evaluate once on unseen 2026 test set, build SHAP plots, and deliver final report.